HIRNA DEREGE

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PROJECTS & OUTSIDE EXPERIENCE

BiliMetrix (Senior Capstone Project) ***September 2025 - June 2026***

- Calculates bilirubin levels in blood from a snapshot of a test strip to diagnose and prevent kernicterus
- Created and optimized detectors to improve readings, updated UI in Summary Page
- Mainly written in C#, learned how to work with MAUI, Xamarin, XML, and about mobile apps

Thread-Safe GPU Allocator (Research Project) ***June 2025 - August 2025***

- Enabled safe concurrent memory allocation across 4,000+ GPU threads through lock-free synchronization
- Sole researcher and developer on faculty-lead research project
- Achieved millions of memory operations per second while preventing data corruption
- Learned lock-free programming, atomic operations, used C++, CUDA, concurrent debugging tools

Live Visualizer ***August2026 - Present***

- Built a browser-based trace visualizer replaying 4,000,000+ real execution events, showing live memory allocation, thread contention, and retry patterns across 128, 512, and 4,096 thread configurations
- Designed custom Canvas/JS rendering engine with typed arrays and requestAnimationFrame for smooth 60fps playback of execution traces

AntHill ***August2026 - Present***

- Built a cross-platform iOS/Android navigtion app in React Native and TypeScript
- Sole developer on personal project
- Implemented live GPS tracking and auto-advancing turn-by-turn instructions using expo-location,calculating real-world distance between coordinates via the Haversine formula to detect step completion
- Built a custom SVG-based elevation profile chart from scratch (no charting library) to visualize climb/descent along each route option

PROFESSIONAL EXPERIENCE

Coding with Kids ***June 2024 - August 2024***
Instructor ***Seattle, WA***

- Instructed over 75 students in programming fundamentals with high student satisfaction
- Mentored beginning coders who successfully completed their first programming projects
- Fostered increased interest in STEM through engaging, hands-on coding instruction
- Classes include-Minecraft modding, Scratch, intro to web development, and intro to programming

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EDUCATION

Seattle University
Bachelor of Science (BS), Computer Science; Minor in Mathematics ***December 2026***
GPA: 3.2

SKILLS

Skills: C++, CUDA, C#, Python, HTML, CSS, R, JavaScript, Java, TypeScript, React Native, .NET, MAUI, Xamarin, XML, Linux, Git, GitHub, Visual Studio Code, IntelliJ IDEA, Eclipse, JetBrains Rider, JetBrains CLion, Canvas API, SVG, Expo Location, GPS tracking, Mobile app development, iOS development, Android development, Game development, Web development, Machine learning, Big data analytics, Database fundamentals, Data structures, Algorithms, Object-oriented programming, GPU programming, Concurrent programming, Lock-free programming, Atomic operations, Thread-safe memory allocation, Memory management, Concurrency debugging, Execution trace visualization, Data visualization, Navigation app development, Geospatial calculations, Haversine formula, Scratch, Minecraft modding, Programming instruction, Mentoring, Teamwork, Organization, Problem solving, Leadership, CodePath Technical Interview Prep